A UNIQUE FACTORIZATION DOMAIN SATISFIES THE ASCENDING CHAIN CONDITION FOR PRINCIPAL IDEALS

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Allowed standard prerequisites (editorial).

- Prime factorization in a unique factorization domain. Used at the start of the counted proof (line 29767). Writes the first element of the chain as a product of prime powers, fixing a finite list of primes.
- Divisibility of associates in a unique factorization domain. Used at proof lines 29768-29769. Forces every later element of the chain to be an associate of a product of the same primes with bounded exponents.

Lemma 0.1. *A UFD satisfies the ascending chain condition for principal ideals.*

Proof. Consider an ascending chain $(a_1) \subset (a_2) \subset (a_3) \subset \ldots$ of principal ideals in R. Write $a_1 = p_1^{e_1} \ldots p_r^{e_r}$ with p_i prime. Then we see that a_n is an associate of $p_1^{c_1} \ldots p_r^{c_r}$ for some $0 \leq c_i \leq e_i$. Since there are only finitely many possibilities we conclude. $\square$

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